Locally Plausible, Globally Wasteful: Consensus Consolidation Can Degrade Shared Experience in Multi-Agent LLM Systems

Anonymous submission

Abstract

Multi-agent LLM systems increasingly share experiential memory: agents write distilled insights into a common pool that is retrieved and injected into every teammate's context. We study a controlled failure mode of one such design that requires no wrong facts, no poisoned text, and no illegal actions. In a trap-maze microscope where every route is legal and shortest-path cost is exactly measurable, we run a preregistered ladder of experiments using implementations based on published components (ExpeL-style insight operations and Generative-Agents-style retrieval). Across a five-arm multi-agent matrix, active memory degrades efficiency and amplifies looping; under our shared-consolidated protocol, the effect is largest relative to the write-only frozen control (success-only excess steps $+11.8$, 95% CI $[7.7, 16.0]$; loop rate $+0.138$, CI $[0.092, 0.188]$; $5/5$ seed signs) and its loop amplification is the only one that significantly exceeds private memory. We do not observe the preregistered read-side importance dose-response; a write-side audit instead finds that consolidation compresses the active strategy supply by $\sim 2.3\times$. The effect persists without self-healing over $T{=}10$, whereas the exploratory $\lambda{=}1$ append arm improves over that horizon. Two archive interventions bound, but do not identify, the mechanism: re-exposure at added dose worsens stagnation, and the equal-dose manipulation does not expand retrieved diversity. A Qwen3-32B probe replicates compression but leaves the efficiency effect unresolved and excludes a loop amplification of the primary model's magnitude. These results identify a bounded risk in a measurable setting rather than a task- or model-universal failure.

1 Introduction

When army ants lose the pheromone trail, each ant follows the locally impeccable rule "follow the ant in front of you," and the colony organizes itself into a circular mill that marches until exhaustion. No single step is wrong; the composition is fatal.

We test whether shared experiential memory in multi-agent LLM systems can admit an analogous failure. The prevailing design pattern—popularized for single agents by experiential learners such as ExpeL (Zhao et al. 2024) and Reflexion (Shinn et al. 2023), and extended to agent teams by experiential co-learning (Qian et al. 2024a) and cross-task multi-agent experience pools (Li et al. 2025)—lets agents distill natural-language insights from their own trajectories, store them in a memory pool, and retrieve them as guidance. When the pool is *shared* and a reviewer LLM *consolidates* it by merging, editing, and up/down-voting insights, every ingredient is individually reasonable and individually published. We test whether their composition has measurable costs.

The failure we test is *silent*. Agents can keep succeeding, never cross walls, and never submit false answers; the experience text is locally plausible and derived from genuine trajectories. Yet routes can grow longer, agents can revisit the same cells, and loop signatures can rise. Because no error is emitted, success alone need not expose this loss.

Studying such a compositional failure on a live benchmark confounds every knob at once. We instead build a *mechanism microscope*: a generated 15×15 trap-maze family with known shortest paths, local observations, and a step cap, in which efficiency loss and looping are exactly measurable per route (Figure 1). Into this microscope we place concrete implementations of published component families (Table 1). This isolates the tested composition—shared memory plus reviewer consolidation—from maze observability and route-cost ambiguity; it does not make the maze result a substitute for deployment evaluation. All confirmatory experiments were preregistered with frozen gates; every deviation and negative result appears in an append-only log.

Our contributions:

- **A preregistered phenomenon in a controlled setting.** In a five-arm memory matrix over five seeds, the shared-consolidated arm degrades success-only route efficiency, amplifies loops, and reduces success relative to a write-only frozen control. Relative to *private* memory, the shared-specific separation is loop amplification, not a broad efficiency gap.
- **A bounded mechanism account.** The preregistered read-side importance dose-response is not supported, while a write-side audit shows a $\sim 2.3\times$ compression of active strategy supply. Over $T{=}10$, the consolidated effect persists while the exploratory $\lambda{=}1$ append arm improves.
- **Intervention boundaries.** Two preregistered archive-

rescue interventions show that added-dose re-exposure does not improve behavior and that an equal-dose manipulation does not expand retrieved diversity. Supply compression remains a robust correlate, not an identified cause.

- **Transfer bounds.** A cross-model probe structurally replicates the write-side compression but not the behavioral coupling, bounding the phenomenon's transfer instead of assuming it.
- **A reusable microscope and discipline.** The anonymized supplementary artifact contains the environment, runners, prompts, configurations, paired-route statistics, sanitizer, memory audits, and preregistrations with freeze hashes.

2 Related Work

Single-agent experiential learning. ExpeL (Zhao et al. 2024) distills insights from success/failure trajectories with ADD/EDIT/UPVOTE/DOWNVOTE operations and retrieves them as guidance; Reflexion (Shinn et al. 2023) converts failures into verbal self-corrections; ReAct (Yao et al. 2023) interleaves reasoning and acting; Generative Agents (Park et al. 2023) score memories by relevance, importance, and recency; MemoryBank (Zhong et al. 2024) adds long-term memory management. These works established that experience helps a *single* agent. Our write operations, retrieval scoring, and distillation prompts follow their component families (Table 1); we do not claim implementation-identical reproductions of the full original systems.

Multi-agent experience sharing. Experiential co-learning (Qian et al. 2024a) shares shortcut experiences between instructor and assistant agents in ChatDev (Qian et al. 2024b); MAEL (Li et al. 2025) equips each agent in a collaboration graph with an experience pool queried by similarity and reward; multi-agent debate (Du et al. 2024) and self-consistency (Wang et al. 2023) aggregate opinions rather than memories; MetaGPT (Hong et al. 2024) shares a message pool. All published sharing designs accumulate experience additively. To our knowledge, no prior work evaluates the natural next step—*LLM-driven consensus consolidation of a shared experience pool*—with controls; that protocol is exactly our subject.

Memory risk and governance. Recent work studies failure modes in LLM memory systems (Garg et al. 2026) and longitudinal risks of memory-equipped agents (Wang et al. 2026). We complement this line with a preregistered, controlled study of one specific protocol—shared reviewer consolidation—with exact route-level ground truth and explicitly reported negative interventions.

3 The Maze Microscope

Environment. We generate a 15×15 "trap" maze family with enforced shortest-path length ≥ 30 and disjoint train/heldout splits per experiment seed. Solvers receive local observations only (no maze graph, no shortest path), act under a 120-step cap, and are scored by exact excess steps over the known optimum. Efficiency loss and loop signatures are therefore measurable per route—the property that makes silent degradation visible at all.

Table 1: Concrete mechanisms adapted from published component families. We retain their operation family or retrieval objective; the shared reviewer update is the evaluated composition, not an implementation-identical reproduction of any full prior system.

Component	Source
Insight ops ADD/EDIT/UP/DOWN, importance-0 deletion	ExpeL
Retrieval scoring (relevance+importance+recency, λ)	Generative Agents
Insight distillation from trajectories	ExpeL
Stepwise solving with injected experience	ReAct-style
Shared pool across agents	Co-learning, MAEL
One reviewer update over a shared pool	**composition under test**

System under test. Four solver agents attempt the *same* tasks in independent environment copies. They never communicate or vote; the *only* coupling channel is the experience memory. Each round t comprises (i) evaluation on 12 held-out mazes per agent (48 routes), (ii) a training batch of 4 mazes per agent (16 routes), and (iii) one reviewer LLM that sees only batched logs and quality signals—never hidden routes—and updates the pool. We use "consensus" operationally for this one shared reviewer update, not for a solver vote or multi-agent dialogue. The final round evaluates without writing. Retrieval injects the top-k=6 insights; the pool is capped at 80.

Protocol detail. In the consolidated arm, the reviewer emits at most six JSON ADD/EDIT/UPVOTE/DOWNVOTE operations over the visible pool. New items begin with two votes; zero-vote items are removed; near-duplicate additions are merged at a fixed similarity threshold. In E2, relevance-only top-k retrieval uses Generative-Agents-style normalization with importance weight λ=0 and recency weight zero. E3 changes only λ. Full prompts, configurations, sanitizer rules, and retry behavior are included in the anonymized supplementary artifact.

Arms. The five-arm matrix decomposes "multi-agent shared consolidated memory" factor by factor: `no-memory` (is multi-agent execution itself pathological?); `frozen` (reviewer writes, solvers never read: is *storage* harmful?); `private` (per-agent pools: is *sharing* necessary?); `shared-append` (shared but never merged: is sharing without consolidation harmful?); and `shared-consolidated` (the protocol under test).

Endpoints and statistics. Primary preregistered endpoints: success-only excess steps, loop rate, and the multi-agent ant-mill rate (loop + route overlap). Auxiliary: success, stagnation, maximum revisits, failure-penalized cost, and route diversity. Mechanism audit, logged per run: pool-size

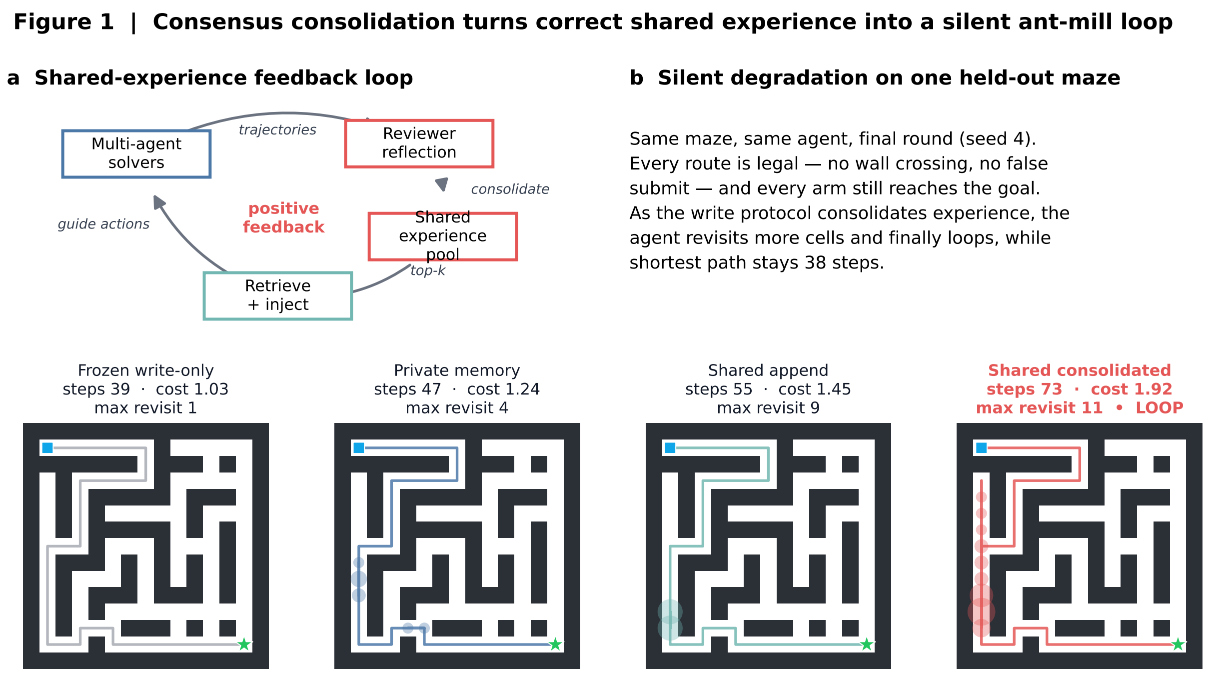

Figure 1: **Mechanism microscope and a silent degradation case.** (a) The shared-experience feedback loop: solver trajectories are reflected into a shared pool, consolidated by LLM-issued operations, retrieved top-k, and injected back as guidance—a positive-feedback cycle. (b) Same held-out maze and agent, final round; shortest path 38. Revisit heat grows as the write protocol consolidates, culminating in a silent loop under shared-consolidated memory, while every route stays legal and successful.

trajectory, *distinct injected experiences*, retrieval entropy and top-1 share, sanitizer rejections, and LLM errors. The same-maze same-agent paired bootstrap (10k resamples, 95% CI) describes paired route variation; it does not treat the 240 routes as 240 independent seed replications. We therefore report per-seed signs alongside pooled intervals. Because excess steps conditions on paired success, we interpret it jointly with success and failure-penalized cost rather than as a standalone population efficiency measure. A sanitizer rejects maze IDs, coordinates, and fixed action sequences from memory text; cache keys are salted by (run, seed, round, task, agent, step). The primary model is DeepSeek-V3 (DeepSeek-AI 2024) via a fixed endpoint; a preregistered drift probe (cache disabled, replaying a frozen-arm round; all probe metrics inside the original bootstrap CIs) authorized reuse of cached controls.

4 Preregistered Results: the Phenomenon

4.1 E1: Single-Agent Attribution Control

Before introducing sharing, we ask whether the implemented experience-learning components change single-agent behavior at all. Relative to a no-memory baseline (paired, final round, three seeds), neither the ExpeL-operations writer nor the append writer moves any primary endpoint: success-only excess steps CIs contain zero (append -4.74, CI $[-14.1, +4.2]$; ops $+5.11$, CI $[-6.4, +16.3]$), as do loop rates. By the frozen rule this is a **fail**, recorded as such. However, the mandated adherence analysis shows experience is *not* ignored: the ops writer shifts action selection away from the controller suggestion after memory activates ($+0.041$ deviation, same sign in $3/3$ seeds), and the append writer significantly reduces maximum revisits. The single-agent null is therefore an *attribution control*: it makes a generic effect of merely enabling memory less plausible, but it does not by itself identify sharing as the causal channel.

Table 2: E2 final-round paired differences versus frozen write-only memory (five seeds, 240 paired routes nested within seeds; success-only excess steps drops unpaired failures). * = paired-route 95% CI excludes zero.

Arm	Succ.	Excess	Loop	Revisit
MAS no memory	-0.004	$+1.12$	$+0.042^*$	$+0.10$
Private	-0.175^*	$+7.32^*$	$+0.079^*$	$+1.20^*$
Shared append	-0.121^*	$+2.75$	$+0.025$	$+0.23$
Shared consol.	-0.171^*	$+11.84^*$	$+0.138^*$	$+1.68^*$

4.2 E2: Multi-Agent Memory Matrix

We compare the five arms at four agents, $T{=}6$, five seeds, against the write-only *frozen* control. Table 2 and Figure 2 report paired differences at the final round.

Three readings follow. First, multi-agent execution is not itself pathological: MAS-no-memory matches frozen on efficiency and success, with only a small loop increase ($+0.042$, CI $[0.008, 0.075]$); frozen itself matches no-memory, so this

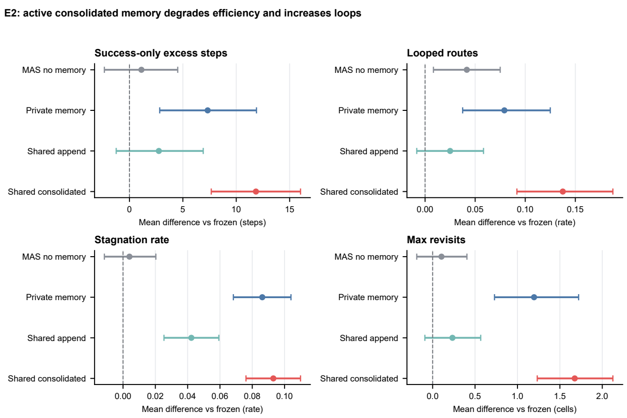

Figure 2: **E2 main result: paired differences versus frozen write-only memory** (five seeds, final round). Points are mean differences; bars are 95% bootstrap CIs; the dashed line is zero. Shared-consolidated (red) degrades on all four endpoints with the largest effects; MAS-no-memory (grey) sits on the zero line except for a small loop increase.

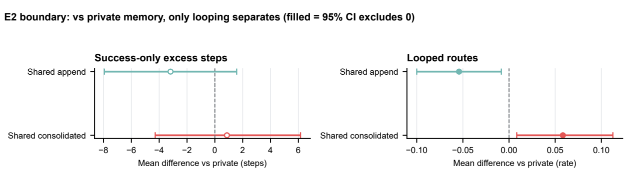

Figure 3: **E2 boundary: shared arms versus a private baseline.** Filled markers mark CIs that exclude zero. Only looping separates shared-consolidated from private ($+0.058$, CI $[0.008, 0.113]$); the efficiency contrast does not, and shared-append loops less than private.

comparison does not reveal a storage-only effect. Second, **within this setting, the shared-consolidated arm meets every preregistered degradation criterion**: success-only excess steps $+11.84$ (CI $[7.66, 16.02]$), same-sign in 5/5 seeds; loop rate $+0.138$ (CI $[0.092, 0.188]$); success -0.171. Third, active injection is a broad risk in this experiment—private memory also degrades—so we test the narrower shared-specific contrast directly.

Shared versus private boundary. Figure 3 contrasts the shared arms against a *private* baseline. The efficiency separation is not significant (consolidated vs. private success-only excess steps $+0.87$, CI $[-4.29, +6.15]$), but loop amplification is: consolidated loops significantly more than private ($+0.058$, CI $[0.008, 0.113]$), while append loops *less*. The honest claim is therefore that consensus consolidation specifically amplifies looping relative to private memory; we do not claim a broad shared-over-private efficiency gap.

4.3 E3: Read-Side Dose-Response Is Null

If the harm were read-side—agents increasingly retrieving a few high-vote memories—then raising the importance weight λ in Generative-Agents-style scoring should deepen degradation. We did not observe that preregistered monotone pattern. Sweeping $\lambda \in \{0, 0.5, 1\}$ on the append arm (three seeds, T=6), the paired difference of λ=1 versus λ=0 on success-only excess steps is $+1.69$ (CI $[-4.26, +7.61]$), loop rate $+0.049$ (CI $[0.000, 0.104]$), and final-round retrieval entropy is non-monotone in λ ($0.833/0.826/0.845$). The read-side usage-weighted-feedback hypothesis is therefore not supported in the tested λ range; this null result is not evidence that every retrieval-side mechanism is harmless.

4.4 Mechanism Audit: Write-Side Strategy-Supply Compression

An offline audit of the same E2 runs locates the concentration on the *write* side (Figure 4). Under identical settings, the consolidated pool converges to 14.2 items with only 11.0 distinct experiences ever injected across all rounds, whereas the append pool grows to the cap (80) and injects 25.4 distinct experiences, and private injects 37.6. Consensus consolidation compresses the population's active strategy supply by roughly $2.3\times$ relative to append. The top-1 retrieval share rises in step, but this is a consequence of the smaller pool rather than an independent channel; the load-bearing evidence is the direct pool-size and distinct-injection counts. Given the intervention results below, we report compression as a robust, cross-checked *correlate* and candidate mechanism, not as established cause.

4.5 E4: Long-Horizon Stress Test (Exploratory)

To ask whether the write-side harm saturates, deepens, or self-heals over a longer feedback horizon, we extend the four-agent shared arms to T=10 (three seeds, heldout 12, frozen and shared-append as controls; not gate-bound, per the preregistration). Relative to frozen, shared-consolidated's degradation is present at every checkpoint tested (t=3, 6, 9) and does not narrow: success drops -0.174, -0.146, -0.257 (all CIs exclude zero) and loop rate rises $+0.083$, $+0.083$, $+0.118$ (all CIs exclude zero), with the final round the worst of the three. A stricter within-condition test—each condition's last three rounds against its first three (excluding warm-up)—finds no significant self-trend for shared-consolidated on any of six endpoints: the gap opens early and persists at a noisy plateau rather than provably deepening. The append arm, in contrast, shows a significant within-condition *improvement* over the same horizon (cost ratio -0.079, excess steps -3.50, stagnation -0.026, max revisits -0.340, all CIs exclude zero) and by t=9 is significantly cheaper than frozen on three endpoints. This within-infrastructure contrast makes a global implementation failure less likely, but it does not establish that append is universally beneficial. Per-seed decomposition exposes heterogeneity the pooled estimate hides: seed 1 shows the only significant late-vs-early decline in success (-0.097); seed 2 has the most severe round-wise trajectory (ending at success 0.17, loop rate 0.46) though its late-vs-early CI includes zero; seed 0 shows no decline. The pooled "no significant further drift" verdict is a mean over a mixed population, not evidence that every instance is equally stable.

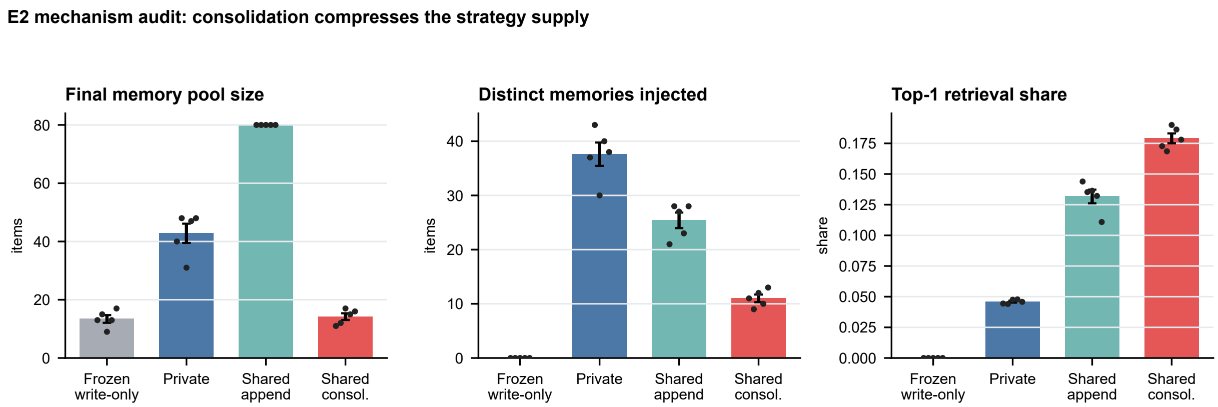

Figure 4: **Mechanism audit: consolidation compresses the strategy supply.** Final pool size, distinct experiences ever injected, and top-1 retrieval share across five seeds (dots). Consensus consolidation keeps a small pool and injects the fewest distinct strategies, concentrating retrieval; the direct counts (left, centre) carry the claim.

Table 3: Preregistered rescue interventions on the consolidated arm (three seeds each). Neither supports promoting supply compression from correlate to cause; both bound the "keep an archive" mitigation.

Arm	Manipulation	Behavioral outcome
Additive archive (top-6 + archive top-6; dose 2×)	supply $\sim$2× (21/21/27 vs. 12/11/14)	no endpoint improves; stagnation worsens $+0.045^*$
Equal-dose joint ranking (total top-6)	supply 1.07–1.55×: **check fails**	not evaluable by frozen rule

4.6 The Failure Is Silent

Figure 1(b) shows a case study mined from the runs: the same held-out maze and agent at the final round, whose shortest path is 38 steps. As the write protocol consolidates, the route degrades monotonically—frozen 39 steps (one revisit), private 47 (four), shared-append 55 (nine), shared-consolidated 73 with cost 1.92, eleven revisits to the same cell, and a detected loop—yet every route is legal and every arm still reaches the goal. This is the ant-mill signature: not a false belief, but a locally plausible correction signal, reinforced through consensus writing, that composes into a globally wasteful cycle.

5 Mechanism Interventions and Boundaries

A second preregistration (Phase Gamma) froze three short, gated extensions: two archive-rescue interventions on the primary model and a cross-model probe (Table 3).

5.1 Rescue I: Additive Archive Access Fails and Backfires

A simple mitigation makes consolidation non-destructive: retain DOWN-voted insights, pre-edit versions, and similarity-merge-discarded variants in a read-only *archive* that solvers can still retrieve. Our first preregistered rescue arm implemented archive access additively (active top-6 *plus* archive top-6), which doubles the injected dose to 12 items per prompt—a preregistered deviation we report explicitly. The manipulation itself succeeded: distinct injected experiences roughly doubled, with archive items winning about half the retrieval slots. Behavior did not improve on a single endpoint: success non-inferiority failed (-0.056, CI $[-0.153, +0.042]$ against a -0.05 bound); success-only excess steps were null (-0.45, CI $[-6.86, +5.96]$); loops trended worse ($+0.035$); stagnation significantly worsened ($+0.045$, CI $[+0.023, +0.067]$). Read as a designed-fix evaluation: naively re-exposing the material consolidation discarded, at added dose, does not rescue behavior and aggravates stagnation—consistent with E2's finding that injection itself is the harm channel.

5.2 Rescue II: the Equal-Dose Archive Is Unreachable

The confounded dose motivated a second, clean arm: one *joint* ranking over active pool and archive with the original total top-6, leaving the per-prompt dose identical to E2. The preregistered manipulation check required the cumulative distinct injected supply to expand by $\geq 1.5\times$ in at least $2/3$ seeds. It did not (ratios $1.42/1.55/1.07$), and not because the archive was ignored—archive items appeared in $896/912$ retrievals where candidates existed. Under a fixed retrieval capacity, consolidated insights dominate the similarity ranking, so the archive substitutes items without expanding *diversity*. Per the frozen amendment, the behavioral gate was not evaluated, and we explicitly do *not* claim that equal-dose archive rescue is ineffective. The constructive reading is that this archive-style design does not restore strategy diversity under our fixed-capacity similarity ranking—a reachability bottleneck that a mitigation must address. Together the two rescue arms leave strategy-supply compression a strong correlate whose causal role is unresolved.

5.3 Cross-Model Probe: Compression Replicates, Coupling Does Not

We repeated the frozen/append/consolidated triangle on Qwen3-32B (Qwen Team 2025) (three seeds, $T{=}4$, justified by E4's finding that the primary-model gap is established by $t{=}3$; thinking mode disabled after a passing calibration smoke: 100% action parse rate, baseline success 0.479). The preregistered gate reports *not-detected-or-underpowered*, but the three endpoints earn three different honest readings (Figure 5):

- *Efficiency: unresolved, direction-consistent.* Success-only excess steps $+5.83$ with all three seeds positive ($+12.4/+0.9/+1.8$), but CI $[-3.75, +15.20]$ ($n{=}40$ success pairs) contains both zero *and* the DeepSeek effect ($+11.84$): the probe cannot adjudicate between no effect and a full-size effect. Power was consumed by Qwen's lower baseline success, which shrinks the success-only pair count.
- *Loops: a bounded difference.* Loop rate -0.076, CI $[-0.153, 0.000]$: the CI *excludes* the primary model's $+0.138$. A loop amplification of that magnitude is absent on Qwen at these settings. We report this as an exploratory magnitude comparison, not a preregistered boundary claim.
- *Success: no drop.* $+0.042$, CI $[-0.049, +0.132]$, excluding the primary model's -0.171.

Crucially, the *mechanism precondition* does transfer: at the matched horizon ($t{=}3$), consolidation compresses the distinct injected supply on Qwen by $1.55\times$ (13.3 vs. 20.7) versus $1.68\times$ on DeepSeek—nearly identical compression, different behavioral coupling. Structural compression without detected harm on one model further cautions against treating compression alone as sufficient cause, consistent with the rescue results above.

6 Design Implications

- **Treat shared consolidation as an audited design choice.** In this setting, the same infrastructure degrades under reviewer consolidation while the exploratory $\lambda{=}1$ append arm improves over a longer horizon. Consolidation should therefore be measured rather than assumed to be hygiene.
- **Monitor the write side in addition to success.** Distinct injected experiences, pool-size trajectory, and retrieval concentration are logged-at-source signals; success alone can miss efficiency decay.
- **Track loops and stagnation in agent telemetry.** Repeated state-action signatures and stagnation separated consolidated from private memory when aggregate efficiency did not.
- **Do not assume an archive fixes consolidation.** Added-dose re-exposure worsened stagnation here, while the equal-dose intervention did not meet its diversity manipulation check. A retained archive must also be retrievable under the deployed ranking rule.

7 Limitations

Our confirmatory evidence is one synthetic environment and one primary model: a controlled maze microscope on DeepSeek-V3. It was chosen because route cost and looping are exactly measurable, not because it establishes deployment behavior. The Qwen probe bounds but does not establish transfer: efficiency is unresolved (underpowered), and the loop signature does not carry at the primary-model magnitude. We include no completed browser or tool-workflow evaluation. The compression mechanism also remains a candidate: the archive interventions constrain it but do not isolate compression from the reviewer update, and we do not compare against deterministic compression or diversity-preserving retrieval baselines. Finally, route-level bootstrap intervals quantify paired trajectory variation, while only five generated splits provide seed-level replication; success-only excess steps is conditional on success and must be read jointly with success and failure-penalized cost. Effects may differ with stronger reviewers, other consolidation prompts, or retrieval schemes beyond similarity ranking.

8 Conclusion

Shared experiential memory warrants measurement as well as capability evaluation. In a preregistered maze microscope, the tested shared reviewer-consolidation protocol can reduce route efficiency, amplify loops relative to private memory, persist over a longer horizon, and compress the active strategy supply. The archive interventions do not identify compression as the cause: added-dose re-exposure worsens stagnation, while the equal-dose intervention does not expand retrieved diversity. The Qwen probe shows that compression can transfer without the observed behavioral coupling. The ant-mill is therefore a bounded warning, not a claim of universal degradation: locally plausible experience can compose into globally wasteful behavior.

Reproducibility. The anonymized supplementary artifact contains preregistrations with SHA-256 freeze records, decision logs, per-route artifacts, memory audits, prompts, configurations, statistics code, and maze runners. Every number in this paper maps to a frozen gate report or an explicitly labeled exploratory analysis.

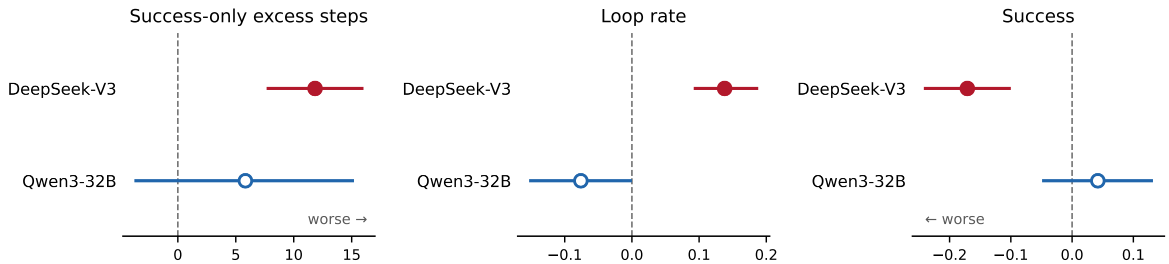

Figure 5: **Cross-model probe: consolidated minus frozen, paired bootstrap 95% CIs.** Filled markers: CI excludes zero. Efficiency (left) is direction-consistent on Qwen but its CI covers both zero and the DeepSeek effect (underpowered); the Qwen loop CI (centre) excludes a DeepSeek-magnitude amplification; success (right) shows no drop on Qwen.

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